TON DUC THANG UNIVERSITY
FACULTY OF ELECTRICAL — ELECTRONICS ENGINEERING & CPAS LAB

OFFICIAL COMPETITION RULES

UAV CONTEST 2027

The 4th International Competition on Research and Design of Unmanned Aerial Vehicles

Celebrating the 30th Anniversary of Ton Duc Thang University (1997 – 2027)

Official Competition Dates: **September 20 and 21, 2027**

Venue: Multi-Purpose Sports Arena, Tan Phong Main Campus, Ho Chi Minh City

Theme: AUTONOMOUS LOW-ALTITUDE LOGISTICS ECOSYSTEM

Version: **Official Rules (Fully Approved)**

All technical parameters and operational criteria officially finalized

Official competition standard issued by the Organizing Committee (BOC) of UAV Contest. Applicable to all participating teams competing on September 20 and 21, 2027.

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I. PURPOSE AND CONTEXT

The low-altitude economy and autonomous logistics are emerging as a strategic digital infrastructure pillar within national science, technology, and innovation development priorities according to Prime Minister Decision No. 808/QD-TTg. Unlike traditional emergency-response scenarios — where a single unmanned aerial vehicle (UAV) operates independently to complete one discrete mission — real-world low-altitude logistics infrastructure operates on a fundamentally different principle: multiple agents (UAVs, ground robots, transfer stations) share the same physical space and must coordinate safely and efficiently, without necessarily relying on a single real-time central controller.

UAV Contest 2027 simulates an **Autonomous Low-Altitude Logistics Distribution Hub**. Each participating team operates a self-built UAV to execute a chain of missions: retrieving packages from a source location, transporting them, and delivering them with precision to a destination point — including a destination that is itself in motion. The competition is organized into three rounds (Preliminary, Semi-Final, and Final) with progressively increasing difficulty and autonomy requirements, alongside a voluntary category — **Coordinated Ops** — which evaluates the ability of two UAVs to coordinate through the standardized CPAS Agent Beacon Protocol (CABP).

II. COMPETITION OBJECTIVES

- **Practical Application:** Developing and optimizing the deployment of UAVs for low-altitude logistics operations — retrieval, transport, and precision delivery within a realistic simulated environment.
- **Innovation and Creativity:** Fostering innovation in airframe mechanics, embedded electronics integration, and advanced flight control under GPS-denied environments.
- **Comprehensive Evaluation:** Concurrently assessing manual piloting skills, fully autonomous capabilities (Edge AI and computer vision), and multi-agent coordination between two UAVs sharing the same physical airspace.

III. UAV TECHNICAL SPECIFICATIONS

1. General Configuration

- **Maximum Dimensions:** $1.0 \times 1.0 \times 1.0$ meters (inclusive of propellers and propeller guards).
- **Maximum Takeoff Weight (MTOW):** ≤ 5.0 kg (including batteries and all onboard mission payloads).
- **Power System:**
  - Battery Type: LiPo/Li-ion, maximum 6S configuration.
  - Quantity: maximum 3 battery packs mounted on the airframe.
  - Total Energy Capacity: ≤ 300 Wh.

- **Payload Capacity:** The UAV is permitted to carry up to five (05) package simulants upon takeoff for the autonomous delivery phase. Packages must be pre-loaded prior to match commencement.
- **System Integrity:** Electrical and mechanical assemblies must maintain high safety standards with zero risk of leakage, short circuits, or thermal runaway.

2. Mandatory Safety Requirements

- **Propeller Guards:** 100% mandatory installation of protective guards/shrouds for all propellers.
- **Manual Override:** Emergency manual control takeover capability must be readily available at all times.
- **Fail-Safe Mechanism:** Automatic activation of Return-to-Home (RTH) or emergency landing upon loss of RC/Telemetry signal exceeding 3 seconds or on critical battery.
- **Status Indicators:** LED status indicators (strongly encouraged).
- **Personnel Safety:** Physical design must present no sharp hazards to operators or referees.
- **Laser Radiation:** Non-ionizing laser systems must comply with IEC 60825-1 (Class 2 / 2M maximum). Ionizing radiation is strictly prohibited.

3. Control Systems

- **Phase 1 — Pickup:** Manual or semi-autonomous flight control permitted.
- **Phase 2 — Delivery:** Fully autonomous flight via waypoints or onboard computer vision. Manual intervention is strictly prohibited, except in emergency situations verified by referees.
- **Coordinated Ops Track:** Both UAVs must continuously broadcast standardized CABP messages at ≥ 5 Hz throughout the entire flight duration.

4. Radio Communication and Localization Equipment

- **Regulatory Compliance:** All radio transmission equipment must strictly comply with Vietnam's Radio Frequency Law.
- **Prohibited Placement:** Placing radio positioning devices (anchors, beacons) directly on the arena playing surface (Playfield) is strictly prohibited.
- **External UWB Permitted:** Positioning systems utilizing anchors located in the perimeter zone outside the playfield (e.g., UWB RTLS) are fully permitted.
- **Signal Disclaimer:** The BOC provides dedicated Wi-Fi channels per team but strongly recommends that teams design backup plans for RF interference.

5. Package Handling Mechanisms

- **Gripping Mechanism:** Must not damage the structural integrity of large package models. Minimum lifting capacity $\geq$ 300g. Must feature a safety locking mechanism preventing unintentional release during flight.
- **Dispensing Mechanism:** Capable of dispensing individual packages with precision. Minimum capacity of 05 packages at takeoff. During dispensing, the lowest point of the UAV must maintain an altitude $\geq$ 3.0 meters above the floor.

6. Standardized Competition Materials

- **Package Simulants (Small):** Standardized to regulation tennis balls, diameter 65.41–68.58 mm, mass 56.0–59.4 g, delivered into Delivery Points D1–D5.
- **Large Package Models:** Rigid cubic block $0.3 \times 0.3 \times 0.3$ meters, mass 300 grams, equipped with an integrated handle or hook for gripping.

IV. ARENA SPECIFICATIONS

- **Overall Dimensions:** 12.0×24.0 meters.
- **Depot (Safe Zone):** 12.0×3.5 meters — houses takeoff/landing pads and mission dispatch/arrival stations.
- **Air Corridor Checkpoints (ACC1 to ACC4):** Inner ring diameter 1.6 meters; center height 1.7 meters; total structure height 2.6 meters. ACC1–ACC2 serve the outbound path; ACC3–ACC4 serve the return path.
- **Pickup Area:** 12.0×4.5 meters. Stations P1, P2, P3 have platform dimensions 1.0×1.0 meters at an elevation of 1.0 meter above the floor.
- **Delivery Points (D1 to D5):** Positioned 0.1 meters above the floor, equipped with integrated sensors for automatic score validation.
- **Operational Altitude Limits:** Manual flight $\leq$ 5.0 meters; Autonomous flight 3.0–5.0 meters.
- **Mobile Receiver (Final Round only):** Moves linearly back-and-forth at a constant velocity of 10 **cm/s** along a fixed guide rail.

V. MISSION REQUIREMENTS

A. PRELIMINARY ROUND

Phase 1 — Pickup (Manual or Semi-Autonomous Control)

1. **Takeoff and Approach:** UAV departs from Depot, passing cleanly through the center of ACC1 and ACC2.

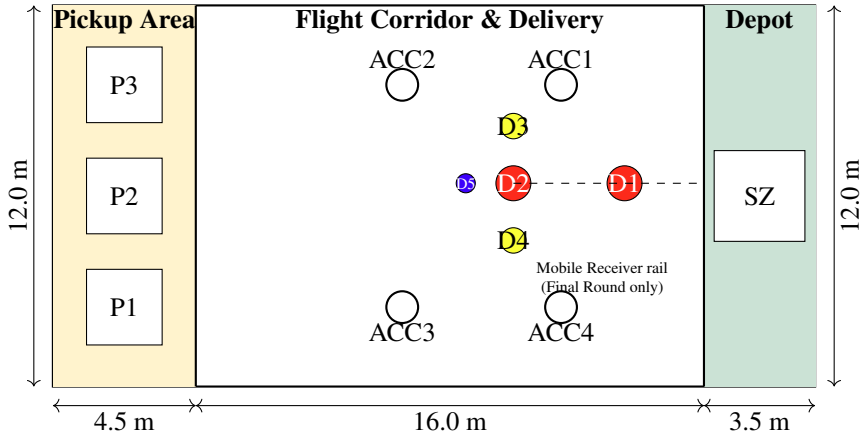


Figure 1: Arena layout schematic (12×24 m). D1,D2: Red (⌀1.0m); D3,D4: Yellow (⌀0.6m); D5: Blue (⌀0.4m).

2. **Pickup Execution:** UAV approaches P1, P2, or P3 and retrieves exactly one (01) large package model.
3. **Transport and Return:** UAV transports the package back through ACC3 and ACC4, placing it safely into the Depot.
4. **Landing:** Controlled landing at the designated pad.

Phase 2 — Delivery (100% Autonomous Flight)

- **Mission 1 — Autonomous Takeoff:** Autonomous mode engaged; UAV climbs to an altitude ≥ 3.0 meters without RC intervention.
- **Mission 2 — Precision Delivery:** UAV navigates to D1–D5 and dispenses package simulants into target rings. In the Preliminary Round, target coordinates are **FIXED**. Each point scores only upon the first valid drop.
- **Mission 3 — Autonomous Precision Landing:** UAV autonomously returns to Depot, performs a precision landing, and disarms motors.

B. SEMI-FINAL ROUND

- Phase 1 procedure is identical to the Preliminary Round.
- Phase 2: Coordinates for D1–D5 are **NOT FIXED**. The Referee Panel conducts a random draw from the position matrix prior to each match. UAVs must use onboard computer vision to identify targets.

C. FINAL ROUND

- Phase 1 procedure is identical to the Preliminary Round.
- Phase 2: Coordinates for D1–D5 are drawn randomly. **Target D2 is mounted on the Mobile Receiver**, moving at a constant speed of 10 cm/s along a fixed linear guide path.

D. VOLUNTARY TRACK: COORDINATED OPS — "PLACE — RELAY — RETRIEVE"

Applies to teams operating **two UAVs**: **UAV-A** (manual/semi-autonomous) and **UAV-B** (fully autonomous) in a sequential relay chain:

Operational Rules and Validation

- **Voluntary Participation:** Independent bonus track. Teams failing or opting out retain 100% of their standard-track score.
- **Candidate Draw:** The Referee Panel draws a candidate subset of exactly **03 out of 05 D-points** (a 3/5 subset of D1–D5).
- **Code Integrity Checksum:** Teams submit the **SHA-256 checksum** and software file copy for UAV-B to the Technical Committee at least 30 minutes prior to the draw.
- **Standardized QR Codes:** Each point D carries a standard 15×15 **cm** QR code affixed at its center.

Phase 1 — Placement (UAV-A) Within time window $T_1 = 90$ **seconds**, UAV-A places 01 package accurately at one of the 3 candidate points. Upon placement, UAV-A transmits the ‘mission_event‘ message via CABP.

Phase 2 — Retrieval (UAV-B) Within time window $T_2 = 120$ **seconds**, measured from Phase 1 completion:

- **Bonus Path (CABP Successful):** If UAV-B receives a valid ‘mission_event‘ within the time-out threshold of 3.0 **seconds**, it skips sequential scanning and flies directly to the destination point.
- **Baseline Path (Automatic Fallback):** If 3.0 seconds elapse without receiving a valid message, UAV-B **automatically initiates** sequential scanning of candidate points via QR code verification and executes retrieval.

Total Match Duration for Coordinated Ops Officially finalized at **07 minutes** (compared to 5 minutes for the standard track).

VI. SCORING CRITERIA

A. Preliminary and Semi-Final Rounds Scoring

Category	Mission Details	Scoring Criteria	Max. Score
Phase 1: Pickup	Air Corridor Checkpoints (ACC1–ACC4)	5 points per checkpoint	20
	Successful retrieval and return to Depot	20 points per package (P1–P3)	60

Category	Mission Details	Scoring Criteria	Max. Score
	Precision landing	10 points	10
		Phase 1 Total	**90**
Phase 2: Delivery	Autonomous takeoff (altitude $\geq$ 3.0m)	20 points	20
	Delivery to D1, D2 (Red, ∅1.0m)	20 points × 2 locations	40
	Delivery to D3, D4 (Yellow, ∅0.6m)	30 points × 2 locations	60
	Delivery to D5 (Blue, ∅0.4m)	50 points × 1 location	50
	Autonomous precision landing at Depot	30 points	30
		Phase 2 Total	**200**
		TOTAL STANDARD SCORE	**290**

B. Final Round Scoring

Category	Mission Details	Scoring Criteria	Max. Score
Phase 1: Pickup	(Identical structure to Preliminary Round)		90
Phase 2: Delivery	Autonomous takeoff	10 points	10
	Delivery to stationary D1 (Red, ∅1.0m)	10 points	10
	Delivery to dynamic D2 — Mobile Receiver (Red, ∅1.0m)	50 points	50
	Delivery to D3, D4 (Yellow, ∅0.6m)	30 points × 2 locations	60
	Delivery to D5 (Blue, ∅0.4m)	50 points	50
	Autonomous precision landing	20 points	20
		Phase 2 Total	**200**
		TOTAL FINAL ROUND SCORE	**290**

C. Bonus Scoring — Coordinated Ops Track (Officially Finalized)

Completion Criterion	Bonus Score
Completing chain via **Bonus Path** (CABP received in 3s, direct flight)	**+60 points**
Completing chain via **Baseline Path** (Sequential QR visual scanning)	**+40 points**

Completion Criterion	Bonus Score
Package placed in T_1, UAV-B departs but fails retrieval in T_2	**+10 points**
Safe separation maintained between both UAVs ($\geq$ 1.5m in CABP logs)	**+10 points**

VII. PENALTIES AND VIOLATIONS

1. Point Deductions

- **Obstacle collision:** Contact with an ACC frame: -5 points per occurrence.
- **Boundary violation:** Flying outside arena boundary > 5 seconds: -5 points per occurrence.
- **Altitude violation:** Flying below 3.0m in Phase 2 > 5 seconds: -5 points per occurrence.
- **Collision between 2 UAVs: Deduct 20 points and terminate the match attempt immediately** for flight safety.

2. Non-Scoring Attempts

- Package dropped prior to reaching Depot floor: 0 points for that package.
- Simulant dropped outside target or duplicate drop into an achieved target: 0 points.

3. Retry Protocol

Maximum one (01) retry per match. The match timer **does not stop**. All points scored prior to the retry are reset to zero.

VIII. MANDATORY SAFETY REGULATIONS

- **Pre-Flight Inspection:** Present UAV 15 minutes before match. Propeller guards 100% mandatory. Damaged or deformed batteries are immediately disqualified.
- **Fail-Safe and Emergency Stop:** Signal loss > 3 seconds triggers automatic RTH or landing. GCS must have a visible emergency kill-switch.
- **Personal Protective Equipment (PPE):** Safety goggles and closed-toe shoes mandatory within the arena.

IX. APPEALS AND DISPUTE RESOLUTION

Appeals must be submitted in writing within **05 minutes** following match completion. Only the official Team Captain may file an appeal. Decisions rendered by the Board of Judges (BOJ) based on CCTV footage and sensor logs are final and binding.

X. TEAM ELIGIBILITY, REGISTRATION, AND DUAL-TRACK FEE STRUCTURE

X.1 Team Composition and Eligibility

- **Eligible Participants:** Undergraduate students, graduate students (Master's and PhD candidates) from domestic and international universities, academic institutes, and research laboratories.
- **Team Roster Quotas:** PhD: max 1; Master's: max 2; Undergrad: max 6; Total official roster capped at **06 members** per team.
- **Arena Personnel (Coordinated Ops):** A maximum of **02 members** (01 pilot for UAV-A and 01 GCS safety technician for UAV-B holding the physical Emergency Stop switch) are permitted inside the technical flight zone simultaneously.

X.2 Dual-Track Fee Structure and Partnership Model

To ensure operational excellence, provide world-class testbed facilities, and foster collaborative academic engagement, the Organizing Committee establishes a dual-track fee structure for **UAV CONTEST 2027**:

1. **Co-Hosting Academic Partner Tier — $5,000 USD / Institution:**
   Designed for partner universities and strategic academic institutes. This contribution entitles the institution to:
   - Official **Co-Organizer status** (co-branding prominently featured across all digital/print media, official backdrops, website, and competition proceedings).
   - One (1) appointed seat on the **International Technical Evaluation Board** and Board of Judges.
   - Complimentary, full-access registration for up to **two (2) official student teams** without any additional entry fees.
   - Priority exhibition booth in the Innovation & Robotics Showcase area during the tournament.
2. **Independent Team Entries:**
   Applicable to individual university teams registering independently:
   - **Domestic Teams (Vietnam): 2,500,000 VND / team** ($\approx$\$100 USD commitment and logistical accreditation fee).
   - **International Teams: $350 USD / team** (or bundled at **$500 USD** for two teams from the same non-partner academic institution).

Registration Tier	Accreditation Fee	Key Entitlements
Co-Hosting Academic Partner	**$5,000 USD /** Institution	Co-Organizer status, branding, 1 seat on Evaluation Board, 2 teams included
Domestic Team (Vietnam)	**2,500,000 VND /** team (≈$100 USD)	1 team entry, full SDKs, field integration
International Team (Single)	**$350 USD /** team	1 team entry, full SDKs, field integration
International Team (Bundle)	**$500 USD /** 2 teams	2 team entries from same institution

Entitlements for All Accredited Teams:
All accredited teams (both Co-Hosting Partner teams and Independent teams) receive:

- Official field and pit area accreditation badges for all registered members and faculty advisors.
- Complete open-source technical Starter Kit SDKs (Python and C++ implementations of the CPAS Agent Beacon Protocol – CABP).
- Pre-match calibration and integration test slots inside the arena equipped with ground-truth Ultra-Wideband (UWB) and Visual-Inertial Odometry (VIO) localization infrastructure.
- Certificate of Participation / Official Tournament Awards and access to technical workshops.

XI. MATCH DURATION AND TIMING

- **Standard Track (Single-UAV):** Maximum **05 minutes** (300 seconds).
- **Coordinated Ops Track:** Maximum **07 minutes** (420 seconds).
- **Setup Time:** 60 seconds allocated prior to match clock initiation.

XII. DETAILED ARENA TECHNICAL SPECIFICATIONS

1. Large Package Model

Rigid cube $0.3 \times 0.3 \times 0.3$ meters, total height with handle 450 mm, weight 300 grams.

2. Delivery Point Standardization

Point	Color	Diameter	HEX Color Code
D1, D2	Red	1000 mm	`#FC2A1B`
D3, D4	Yellow	600 mm	`#FAFF37`
D5	Blue	400 mm	`#3C00FB`

3. QR Codes for Coordinated Ops

Each point D1–D5 features a standard 15×15 **cm** QR code affixed at its center encoding location ID (e.g., `"D1"`, `"D2"`).

4. Competition Materials

Package simulants use regulation tennis balls (∅65.41–68.58 mm, mass 56.0–59.4 g).

XIII. STANDARDIZED CABP PROTOCOL SPECIFICATIONS

The CPAS Agent Beacon Protocol (CABP) operates via UDP Broadcast on default port `14555`:

1. Two Mandatory Message Types

(a) Periodic State Beacon (5 Hz) Broadcast continuously at 200 ms intervals. Binary payload of 31 bytes:

Field	Data Type	Description
`agent_id`	uint16	Unique UAV identifier in match
`seq_num`	uint32	Monotonically increasing sequence number
`agent_state`	enum (uint8)	IDLE (0), EXECUTING (1), RETURNING (2), LANDING (3), EMERGENCY (4)
`intent`	enum (uint8)	HOVER (0), GO_TO_POINT (1), PICKUP (2), DELIVER (3), RETURN_HOME (4), LAND (5)
`timestamp`	uint64	Epoch timestamp (ms)
`position (x,y,z)`	float32 × 3	Arena Cartesian coordinates (meters)
`battery_critical`	bool	Critical battery flag (0 or 1)

(b) Discrete Event Message (`mission_event`) Transmitted 3 times upon successful package placement by UAV-A. Binary payload of 16 bytes:

Field	Data Type	Description
`agent_id`	uint16	Source agent (UAV-A)
`event_type`	enum (uint8)	1 = `PACKAGE_PLACED`
`location_id`	uint8	Target ID (1=D1, 2=D2, 3=D3, 4=D4, 5=D5)
`timestamp`	uint64	Event timestamp (ms)

2. Standardized Timeout Thresholds

- Timeout threshold for `mission_event`: 3.0 **seconds** (subsequently triggering fallback).
- Loss of periodic beacon connection threshold: 500 **milliseconds**.

3. Authority Monotonicity Principle

CABP messages provide advisory guidance only. Final motor actuation authority rests strictly with the Flight Controller. Network packet loss must never compromise aircraft control stability.

XIV. APPENDIX: OFFICIALLY APPROVED TECHNICAL SPECIFICATIONS

Summary of all 11 finalized technical parameters approved by the Organizing Committee for the competition on September 20 and 21, 2027:

No.	Technical Parameter	Officially Approved Specification
1	Candidate subset size for draw	Exactly **03 out of 05 points** (3/5 D1–D5)
2	UAV-B software verification	Submit **SHA-256** hash 30 minutes prior to draw
3	Phase 1 placement window (T_1)	**90 seconds**
4	Phase 2 retrieval window (T_2)	**120 seconds**
5	Timeout threshold for `mission_event`	**3.0 seconds** (automatic fallback to QR scanning)
6	Coordinated Ops match duration	**07 minutes** (420 seconds)
7	Coordinated Ops bonus scoring	**+60 pts** (Bonus) / **+40 pts** (Baseline) / **+10 pts** (Partial) / **+10 pts** (Safe separation)
8	Penalty for inter-UAV collision	**Deduct 20 points and immediate match termination**
9	Arena personnel quota (Coordinated Ops)	Max **02 members** (01 pilot + 01 GCS technician with E-Stop)
10	QR code physical dimensions	Standardized to 15×15 **cm**
11	CABP periodic beacon frequency	Minimum **5 Hz**; Loss detection timeout **500 ms**

Organizing Committee of UAV Contest 2027 — CPAS Laboratory, Ton Duc Thang University.